

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent models, BERT is designed to pre-train deep bidirectional representations from unlabeled text.

The Masked Language Model (MLM): In order to train a deep bidirectional representation, we randomly mask 15% of the input tokens at random, and then predict those masked tokens.

Next Sentence Prediction (NSP): Many important downstream tasks such as Question Answering and Natural Language Inference are based on understanding the relationship between two sentences. We pre-train for a binarized next sentence prediction task.

BERT architecture is a multi-layer bidirectional Transformer encoder based on the original Transformer implementation described in Vaswani et al.

On the GLUE benchmark, BERT achieves an overall score of 80.5%, obtaining substantial improvements across all nine individual tasks including MNLI, QQP, and SST-2.

Fine-tuning BERT on SQuAD v1.1 question answering achieves 93.2 F1 score, outperforming human performance on the benchmark.